s4_tilted_size_neutral_aggressive

equity strategy | family: equity-aggressive | status: research | derived from `s4_tilted_size_neutral`

What this strategy does

s4_tilted_size_neutral re-sized for return: base unit 0.007, 6x leverage cap

Why it should work. Concentrate capital where the surprise is largest; neutralize size so it is not a hidden bet.

Design notes (from the strategy's source file).

Aggressive-sizing variants of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.007
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

metric	s4 tilted size neutral aggressive	market (CRSP VW)
Annualized return	10.70%	9.94%
Annualized volatility	12.25%	19.35%
Sharpe (NAV-pct)	0.89	0.59
Sortino	1.27	
Max drawdown	-40.0%	-55.5%
Calmar	0.27	
Final NAV ($10M start)	$72,371,948	
Worst day	-6.54%	
% positive days	51.9%	
Sharpe 1996-2001	1.30	

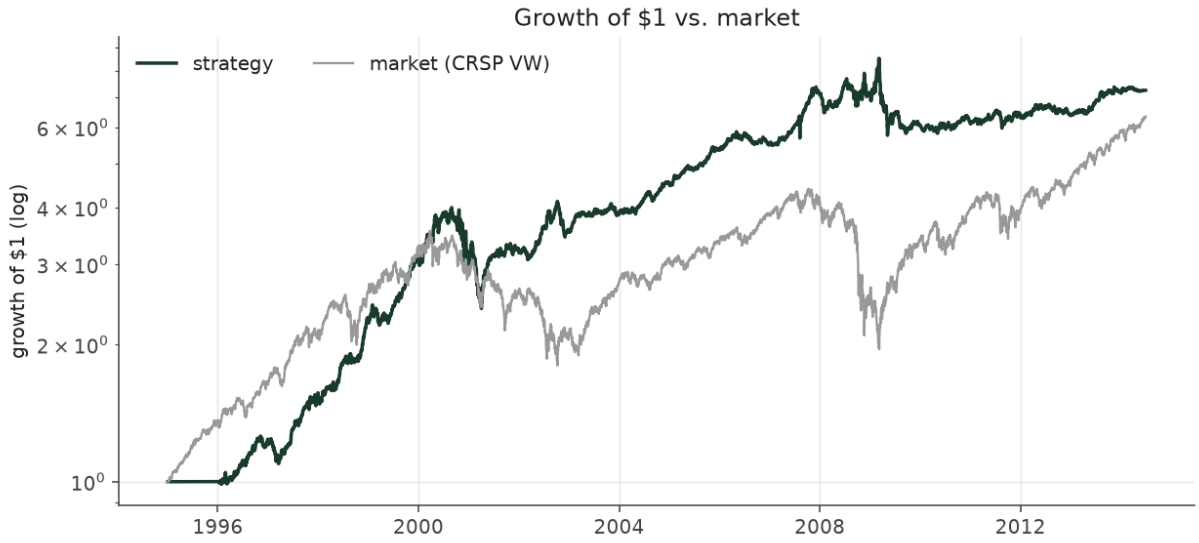
Sharpe 2002-2007	1.54	
Sharpe 2008-2013	0.07	

Correlation of daily returns with the market: **0.03**.

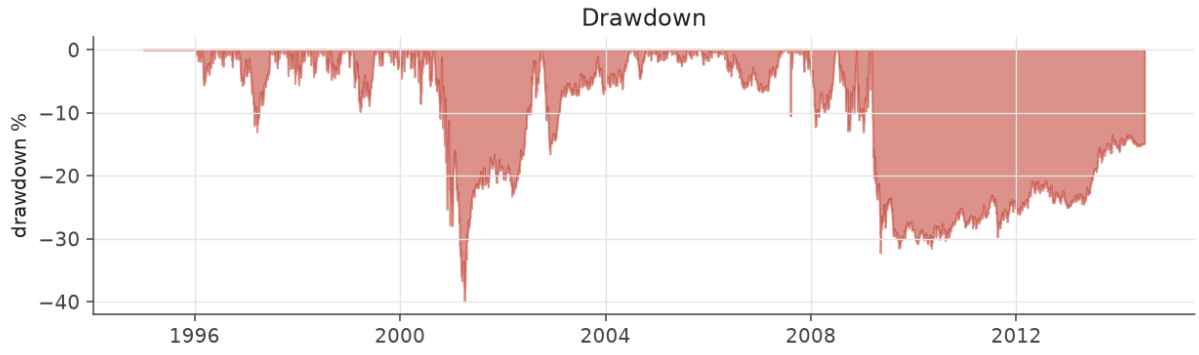
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 0.891

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.107

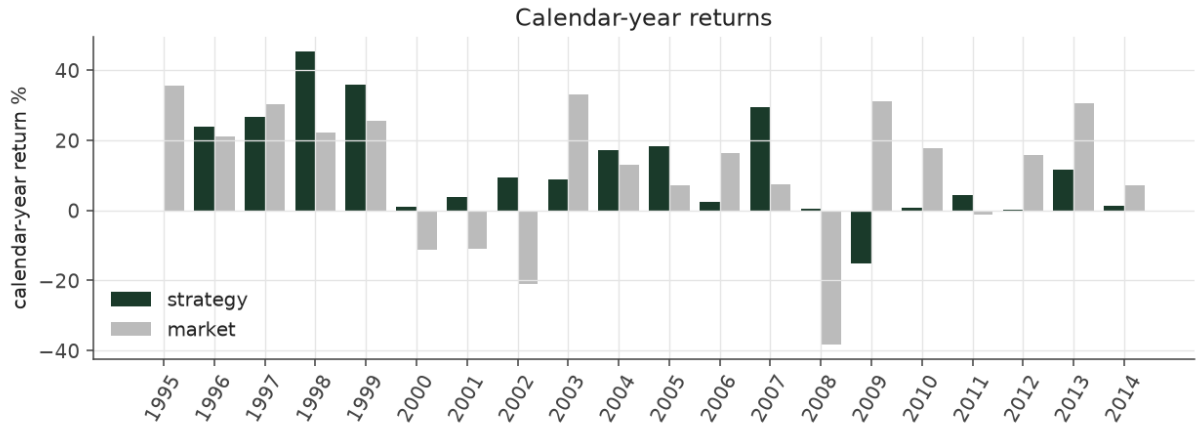
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.267



NAV (log scale)



Drawdown from running peak

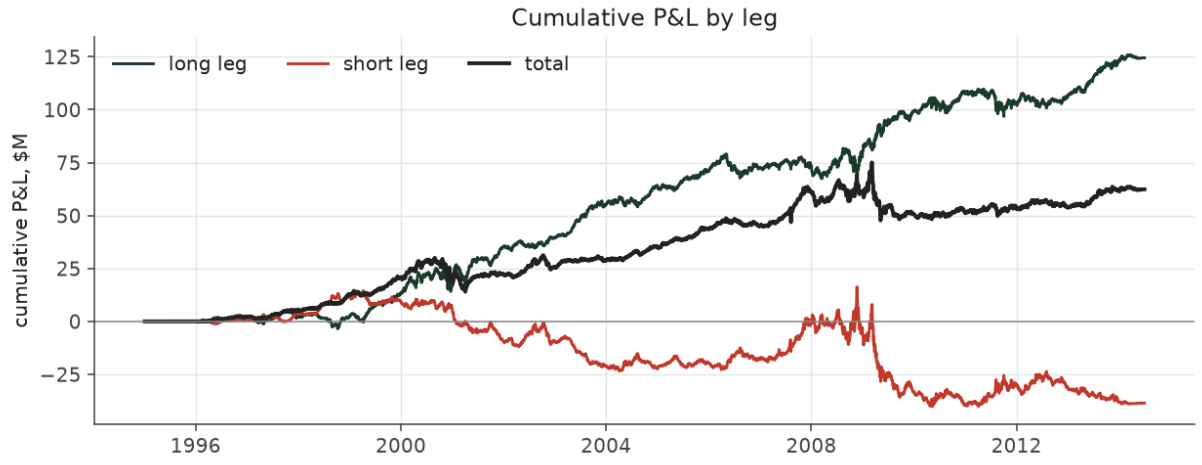


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	124.48	-38.59
P&L after costs ($M)	113.18	-50.81
of which common (average-event) component ($M)	105.32	-105.36

of which excess / PEAD component ($M)	19.16	66.77
excess minus costs ($M)	7.87	54.55
gross return per $ traded (bp)	189.32	-56.80
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	0.95	-0.15

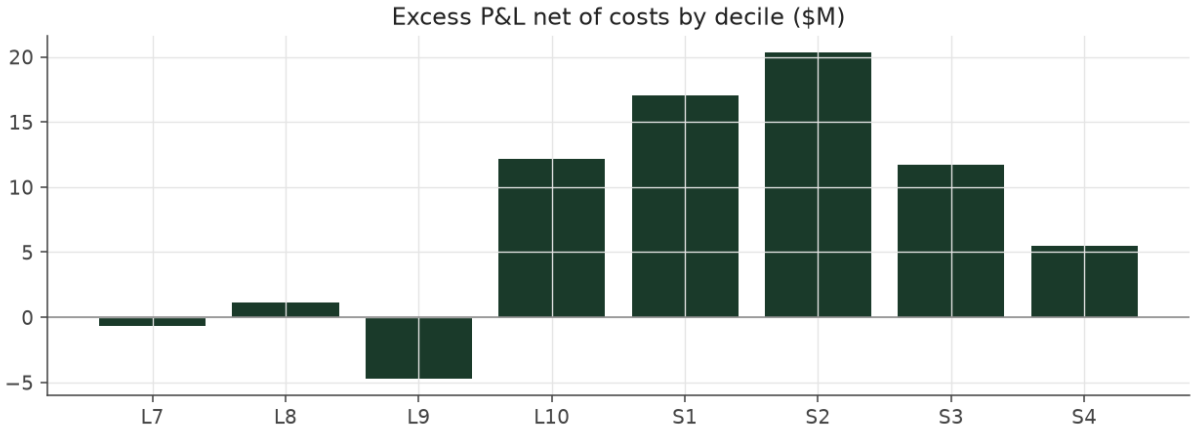
Daily P&L correlation between the two legs: **-0.72**.

• **The short leg loses money in absolute terms** (-38.6M before costs) while the long leg earns +124.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 105.3M / 105.4M. Stripping that common component, the surprise signal itself earned +19.2M on the long side and +66.8M on the short side.

• The two legs' daily P&L are -0.72 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.95 long / -0.15 short vs 0.89 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	531.28	8.56	0.29	0.97	-0.68	-1.1%	-12.76
long	8	16,588	1,498.23	26.40	3.80	2.65	1.15	1.8%	7.65
long	9	17,747	2,072.79	26.15	-1.23	3.50	-4.73	-7.6%	-22.83
long	10	17,174	2,472.81	52.07	16.30	4.17	12.13	19.4%	49.05
short	1	17,169	2,422.99	-18.52	21.39	4.38	17.01	27.2%	70.19
short	2	17,281	2,130.37	-14.14	24.11	3.77	20.34	32.6%	95.46
short	3	15,970	1,602.41	-13.84	14.65	2.92	11.73	18.8%	73.20
short	4	9,171	638.34	-4.31	6.63	1.15	5.48	8.8%	85.79



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	602.96	4.85	8.74	2.02	6.72	10.8%	111.47
2	22,885	2,160.87	17.58	23.79	5.48	18.31	29.3%	84.73

3	25,501	3,273.89	21.32	26.08	6.45	19.62	31.4%	59.94
4	27,583	3,629.42	11.28	16.02	5.50	10.52	16.9%	28.98
5	28,947	3,702.08	7.34	11.31	4.06	7.25	11.6%	19.57

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	301.82	5.97	8.05	1.51	6.54	10.5%	216.62
2	23,084	1,899.67	12.57	20.69	5.70	14.99	24.0%	78.91
3	25,851	3,614.33	23.26	31.93	7.23	24.70	39.6%	68.33
4	27,665	3,809.07	10.53	15.12	5.33	9.78	15.7%	25.68
5	28,874	3,743.30	10.05	10.13	3.74	6.39	10.2%	17.07

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	5,376.99	-6.32	24.09	8.71	15.38	24.6%	28.61
41-60d	25,652	2,972.62	15.99	14.30	5.33	8.97	14.4%	30.19
61-90d	35,324	3,914.33	19.92	19.96	7.17	12.79	20.5%	32.68
>90d	12,298	1,105.28	32.79	27.57	2.30	25.27	40.5%	228.63

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	2,125.65	21.61	24.64	3.67	20.97	33.6%	98.66
2002-07	39,956	4,622.80	35.24	41.74	8.03	33.71	54.0%	72.91
2008-13	45,278	6,620.77	5.52	19.55	11.81	7.74	12.4%	11.69

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	2,356.09	15.94	9.07	4.12	4.95	7.9%	21.00
Chems	2,590	305.15	1.28	2.20	0.51	1.69	2.7%	55.42
Durbl	2,843	306.15	0.81	2.26	0.55	1.71	2.7%	55.79
Enrgy	6,194	728.42	8.18	10.62	1.19	9.43	15.1%	129.48
Hlth	11,696	1,213.37	10.05	8.32	2.27	6.05	9.7%	49.90
Manuf	12,034	1,336.21	2.73	7.14	2.32	4.82	7.7%	36.09
Money	17,583	2,118.09	7.17	19.32	3.69	15.63	25.0%	73.80
NoDur	5,081	574.32	2.52	2.98	0.98	2.00	3.2%	34.88
Other	18,377	2,048.37	7.91	12.23	3.86	8.38	13.4%	40.90
Shops	11,984	1,335.59	7.65	8.84	2.35	6.49	10.4%	48.61

Telcm	3,981	448.63	-1.34	1.22	0.73	0.49	0.8%	10.84
Utils	4,672	598.84	-0.53	1.72	0.95	0.77	1.2%	12.85

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	4,710.15	10.53	19.83	7.61	12.22	19.6%	25.94
low vol	40,091	4,230.51	24.63	32.68	8.07	24.61	39.4%	58.16
mid vol	37,903	4,264.38	26.18	32.44	7.52	24.92	39.9%	58.45

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	10.70%	0.00%
0.5%	6.7	10.15%	0.55%
1.0%	13.3	9.55%	1.15%
2.0%	26.6	8.12%	2.58%
5.0%	66.6	-2.80%	13.50%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.